

Spring 1977

Problem 1 (Sp77, Su82, Fa18, Fa21). Suppose f is a differentiable real valued function such that $f'(x) > f(x)$ for all $x \in \mathbb{R}$ and $f(0) = 0$. Prove that $f(x) > 0$ for all positive x .

Problem 2 (Sp77). Suppose that f is a real valued function of one real variable such that

$$\lim_{x \rightarrow c} f(x)$$

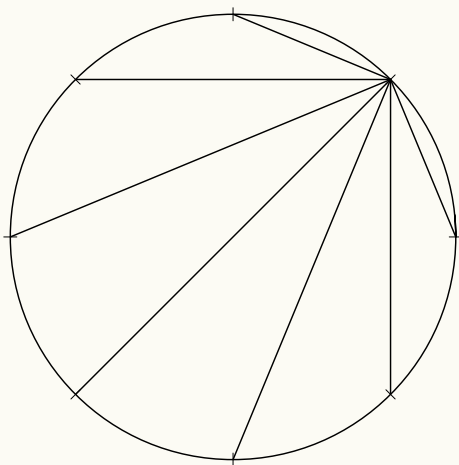
exists for all $c \in [a, b]$. Show that f is Riemann integrable on $[a, b]$.

Problem 3 (Sp77). 1. Evaluate $P_{n-1}(1)$, where $P_{n-1}(x)$ is the polynomial

$$P_{n-1}(x) = \frac{x^n - 1}{x - 1}$$

2. Consider a circle of radius 1, and let $Q_1, Q_2, \dots, Q_n$ be the vertices of a regular n -gon inscribed in the circle. Join Q_1 to $Q_2, Q_3, \dots, Q_n$ by segments of a straight line. You obtain $(n - 1)$ segments of lengths $\lambda_2, \lambda_3, \dots, \lambda_n$. Show that

$$\prod_{i=2}^n \lambda_i = n$$



Problem 4 (Sp77, Sp82, Sp20). Prove the Fundamental Theorem of Algebra: Every nonconstant polynomial with complex coefficients has a complex zero.

Problem 5 (Sp77, Sp78). Let $\mathbb{F} \subset \mathbb{K}$ be fields, and a and b elements of $\mathbb{K}$ which are algebraic over $\mathbb{F}$. Show that $a + b$ is algebraic over $\mathbb{F}$.

Problem 6 (Sp77). Let G be the collection of 2×2 real matrices with nonzero determinant. Define the product of two elements in G as the usual matrix product.

1. Show that G is a group.
2. Find the *center* Z of G ; that is, the set of all elements z of G such that $az = za$ for all $a \in G$.
3. Show that the set O of real orthogonal matrices is a subgroup of G . Show by example that O is not a normal subgroup.
4. Find a nontrivial homomorphism from G onto an abelian group.

Problem 7 (Sp77). A matrix of the form

$$\begin{pmatrix} 1 & a_0 & a_0^2 & a_0^n \\ 1 & a_1 & a_1^2 & a_1^n \\ \vdots & \vdots & \vdots & \vdots \\ 1 & a_n & a_n^2 & a_n^n \end{pmatrix}$$

where the a_i are complex numbers, is called a Vandermonde matrix.

1. Prove that the Vandermonde matrix is invertible if $a_0, a_1, \dots, a_n$ are all different.
2. If $a_0, a_1, \dots, a_n$ are all different, and $b_0, b_1, \dots, b_n$ are complex numbers, prove that there is a unique polynomial f of degree n with complex coefficients such that $f(a_0) = b_0, f(a_1) = b_1, \dots, f(a_n) = b_n$.

Problem 8 (Sp77, Sp93, Fa94). Find a list of real matrices, as long as possible, such that

- the characteristic polynomial of each matrix is $(x - 1)^5(x + 1)$,
- the minimal polynomial of each matrix is $(x - 1)^2(x + 1)$,
- no two matrices in the list are similar to each other.

Problem 9 (Sp77). Find the solution of the differential equation

$$y'' - 2y' + y = 0$$

subject to the conditions $y(0) = y'(0) = 1$.

Problem 10 (Sp77). In $\mathbb{R}^2$, consider the region A defined by $x^2 + y^2 > 1$. Find differentiable real valued functions u and v on A such that $\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$ but there is no real valued function f on A such that $u = \frac{\partial f}{\partial x}$ and $v = \frac{\partial f}{\partial y}$.

Problem 11 (Sp77, Sp25). Let a_n be defined by

$$1 - x^2 + x^4 - x^6 + \cdots = \sum_{n=0}^{\infty} a_n (x+2)^n \quad \text{for } 0 < x < 1$$

Find $\limsup_{n \rightarrow \infty} |a_n|^{1/n}$.

Problem 12 (Sp77). Let p be an odd prime. Let $Q(p)$ be the set of integers a , $0 \leq a \leq p-1$, for which the congruence

$$x^2 \equiv a \pmod{p}$$

has a solution. Show that $Q(p)$ has cardinality $(p+1)/2$.

Problem 13 (Sp77). Consider the family of square matrices $A(\theta)$ defined by the solution of the matrix differential equation

$$\frac{dA(\theta)}{d\theta} = BA(\theta)$$

with the initial condition $A(0) = I$, where B is a fixed square matrix.

1. Find a property of B which is necessary and sufficient for $A(\theta)$ to be *orthogonal* for all θ .
2. Find the matrices $A(\theta)$ corresponding to

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

and give a geometric interpretation.

Problem 14 (Sp77, Su82, Sp24). A square matrix A is nilpotent if $A^k = 0$ for some positive integer k .

1. If A and B are nilpotent, is $A + B$ nilpotent?
2. Prove: If A and B are nilpotent matrices and $AB = BA$, then $A + B$ is nilpotent.
3. Prove: If A is nilpotent then $I + A$ and $I - A$ are invertible.

Problem 15 (Sp77). Let $f(z)$ be a nonconstant meromorphic function. A complex number w is called a period of f if $f(z + w) = f(z)$ for all z .

1. Show that if w_1 and w_2 are periods, so are $n_1 w_1 + n_2 w_2$ for all integers n_1 and n_2 .
2. Show that there are, at most, a finite number of periods of f in any bounded region of the complex plane.

Problem 16 (Sp77, Fa81, Sp82, Sp26). Evaluate $\int_{-\infty}^{\infty} \frac{\cos nx}{x^4 + 1} dx$.

Problem 17 (Sp77). Let $\mathbb{Q}_+$ be the multiplicative group of positive rational numbers.

1. Is $\mathbb{Q}_+$ torsion free?
2. Is $\mathbb{Q}_+$ free?
3. Is $\mathbb{Q}_+$ free abelian?

Note: See also ??????.

Problem 18 (Sp77). 1. In $\mathbb{R}[x]$, consider the set of polynomials $f(x)$ for which $f(2) = f'(2) = f''(2) = 0$. Prove that this set forms an ideal and find its monic generator.

2. Do the polynomials such that $f(2) = 0$ and $f'(3) = 0$ form an ideal?

Problem 19 (Sp77). Suppose that $u(x, t)$ is a continuous function of the real variables x and t with continuous second partial derivatives. Suppose that u and its first partial derivatives are periodic in x with period 1, and that

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$$

Prove that

$$E(t) = \frac{1}{2} \int_0^1 \left(\left(\frac{\partial u}{\partial t} \right)^2 + \left(\frac{\partial u}{\partial x} \right)^2 \right) dx$$

is a constant independent of t .

Problem 20 (Sp77). Let $h : [0, 1) \rightarrow \mathbb{R}$ be a function defined on the half-open interval $[0, 1)$. Prove that if h is uniformly continuous, there exists a unique continuous function $g : [0, 1] \rightarrow \mathbb{R}$ such that $g(x) = h(x)$ for all $x \in [0, 1)$.